

Maneesh Kumar Singh

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RESEARCH INTERESTS

Bayesian Scientific Computing, Data Assimilation, Numerical Analysis, Computational Mathematics.

EMPLOYMENT

- 2022– Imperial College London
Postdoctoral Research Associate, Department of Mathematics
- 2021–22 University of Konstanz
Postdoc Fellow, Department of Mathematics and Statistics
- 2019–21 Indian Institute of Science
Postdoc Fellow, Department of Computational and Data Sciences

EDUCATION

- 2013–19 Ph.D. in Mathematics, Indian Institute of Technology Guwahati
• Thesis title: Analysis of robust computational techniques for singularly perturbed system of parabolic partial differential equations • Supervisor: Prof. Natesan Srinivasan.
- 2011–13 MSc in Applied Mathematics with 7.86 CGPA, University of Hyderabad
- 2008–11 BA (Hons) in Mathematics with 68.89 % Banaras Hindu University

FUNDING AND SCHOLARSHIPS

- 2021 D.S. Kothari Postdoctoral Fellowship, UGC, India
- 2018 NBHM Postdoctoral fellowship, NBHM, DAE, India
- 2013 CSIR-UGC JRF, Gate Fellowship in Mathematics, MHRD, India

TEACHING EXPERIENCE

IIT Guwahati – Teaching Assistant

- 2015–16 Calculus, Linear Algebra & ODE
- 2016–17 Scientific Computing, Linear Algebra & ODE
- 2017–18 Complex analysis & PDE, Computational Finance

Ashoka University – Teaching Fellow

- 2018 Calculus

Imperial College London – Substitute Instructor

- 2022–25 Computational Linear Algebra, Finite Elements

PUBLICATIONS

Journal Articles

Working Articles

M. K. Singh, C.J. Cotter, D. Crisan: Data Assimilation for a Quasi-Geostrophic equation using advanced particle filtering.

D.D. Holm, M. K. Singh, C.J. Cotter: Energy conserving compatible FEM discretisations for deterministic and stochastic rotating shallow water equations.

Submitted Articles

- 2025 M. K. Singh, D. Crisan, J Hope-Collins, C.J. Cotter: Data assimilation using a global Girsanov nudged particle filter (submitted to *Foundation of Data Science*)
- 2025 D.D. Holm, M.K. Singh, O. D. Street: A comparative numerical study of stochastic Hamiltonian Camassa-Holm equations. (submitted to *Proceedings of the Royal Society A*) *arXiv preprint arXiv:2507.14264*
- 2025 D.D. Holm, M.K. Singh: Multiscale geodesic flows. (submitted to *Physics Review Letters*) *arXiv preprint arXiv:2507.05018*

Published Articles

- 2024 S. Frej, M. K. Singh: An implicitly extended Crank-Nicolson scheme for the heat equation on a time-dependent domain. *Journal of Scientific Computing*, 99(3) 64: 1-29
- 2024 S. Ganesan, M. K. Singh: Operator-splitting finite element method for solving the radiative transfer equation. *Numerical Algorithms*, 98: 1725-1753
- 2024 M. K. Singh: A discrete-ordinate weak Galerkin method for radiative transfer equation. *Applied Numerical Mathematics*, 201: 628-641
- 2021 M. K. Singh, G. Singh, S. Natesan: A unified study on superconvergence analysis of Galerkin FEM for singularly perturbed systems of multiscale nature. *Journal of Applied Mathematics and Computing*, 66: 221-243
- 2020 S. Natesan, M.K. Singh: Robust computational method for singularly perturbed system of parabolic convection-diffusion problems with interior layers. *Computational and Mathematical Methods*, 3: e1146
- 2020 M. K. Singh, S. Natesan. Numerical solution of 2D singularly perturbed reaction-diffusion system with multiple scales. *Computers and Mathematics with Applications*, 80(4): 36-53, <https://doi.org/10.1016/j.camwa.2020.04.028>
- 2019 M. K. Singh, S. Natesan: A parameter-uniform hybrid finite difference scheme for singularly perturbed system of parabolic convection-diffusion problems. *International Journal of Computer Mathematics*, 97(4): 275-305
- 2019 M. K. Singh and S. Natesan. Numerical analysis of singularly perturbed system of parabolic convection-diffusion problem with regular boundary layers. *Differential Equations and Dynamical Systems*, 30(3):695-717
- 2019 M. K. Singh, S. Natesan: A robust computational method for singularly perturbed system of 2D parabolic convection-diffusion problems. *International Journal of Mathematical*

- Modelling and Numerical Optimisation*, 9(2): 127-157
- 2018 M. K. Singh, S. Natesan: Richardson extrapolation technique for singularly perturbed system of parabolic partial differential equations with exponential boundary layers. *Applied Mathematics and Computation*, 333: 254-275, <https://doi.org/10.1016/j.amc.2018.03.059>

Book Chapter

- 2024 C.J. Cotter, D. Crisan, M. K. Singh: Data assimilation for the stochastic Camassa-Holm equation using particle filtering: a numerical investigation. *Stochastic Transport in Upper Ocean Dynamics III, Mathematics of Planet Earth*, 13: 137-160

Conference Proceedings

- 2024 C. Cotter, D. Crisan, M. K. Singh: The GONG Particle Filter: A Novel Framework for Data Assimilation in Stochastic PDEs, *CASML 2024, IISc Bangalore*
- 2018 M. K. Singh, S. Natesan: A finite element super convergence approximations for singularly perturbed system of convection-diffusion problems. *AIP Conference Proceedings, FIAM 2018, 1975, 030016*

PRESENTATION

Talks at Conferences

- 2024 Data Assimilation for Stochastic PDEs: The GONG Particle Filter Framework. *5th STUOD Annual Workshop (INRIA, Renne, France)*
- 2024 Advanced particle filtering data assimilation framework for Stochastic PDEs. *Firedrake meeting 2024 (University of Oxford)*
- 2023 Parallelisation schemes for particle filtering data assimilation. *4th STUOD Annual Workshop (IFREMER, France)*
- 2023 Data assimilation algorithm using advanced particle filters for stochastic PDEs. *Firedrake meeting 2023 (London)*
- 2021 Superconvergence study of Galerkin fem for singularly perturbed systems with multiple scales *International Conference ADENA, IIT Guwahati*
- 2018 A finite element super convergence approximations for singularly perturbed system of convection-diffusion problems. *International Conference FIAM, NIT Hamirpur*
- 2017 Uniform superconvergence of a Galerkin Finite element method for singularly perturbed system of reaction- diffusion problems. *62nd Congress OF ISTAM (OU, Hyderabad)*
- 2016 Analysis of an upwind difference scheme on Shishkin-type mesh for singularly perturbed system of convection-diffusion equations. *ICMMDESCA-2016 (IIT Kanpur)*
- 2016 Analysis of a uniformly convergent hybrid numerical method for singularly perturbed system of parabolic problems. *ICTIMCAMS (BHU, Varanasi)*
- 2015 Numerical methods for singularly perturbed system of parabolic partial differential equations. *ICCTP: TC (SAU, New Delhi)*

Talks at University seminars

- 2024 Data assimilation with advanced particle filters for stochastic partial differential equations. *TIFR-CAM Seminar Talk*

- 2024 The Hitchhiker's Guide to FEM: A Beginner's Perspective. *IIIT Gwalior Workshop on PDE*
- 2023 An implicitly extended Crank-Nicolson scheme for the heat equation on time-dependent domains. *ICL/UCL Numerics Seminar*
- 2023 Towards nudging particle filter for Stochastic PDEs. *24th STUOD Sandbox Workshop (ICL)*
- 2022 An Introduction to the Cut Finite Element method (CutFEM). *Research Scholar Seminar (IIT Guwahati)*
- 2021 An operator-splitting finite element method for radiative transfer equation. *Oberseminar Numerik (University of Konstanz)*
- 2019 Five days workshop on parallel finite element computing using ParMooN. *(IISc)*.

ADDITIONAL INFORMATION

- Scientific Computing package: NumPy, SciPy, PyTorch, TensorFlow, ParaView
- Computing environment: FEniCS, Firedrake, MATLAB, deal.II, ParMooN

REFERENCES

- 1 Prof. Natesan Srinivasan(PhD Supervisor), Department of Mathematics, IIT Guwahati, India
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- 2 Prof. Colin Cotter(Postdoc mentor), Department of Mathematics, Imperial College London, UK
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- 3 Prof. Darryl D Holm(Postdoc mentor), Department of Mathematics, Imperial College London, UK
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- 4 Dr. Stefan Frei(Postdoc mentor), Department of Mathematics and Statistics, University of Konstanz, Germany
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